Appendix

Sketch of proof (*3.2.1*). By strong duality, $C_f^{oper}(P) = max_{\lambda_f \geq 0, \varsigma_f} min_{x_f \geq 0} \mathcal{L}_f(x_f, \lambda_f, \varsigma_f; P)$. Danskin's theorem (Danskin., 1966; Al-Dujaili et al., 2019) for pointwise maxima of affine functions in the parameter gives $\partial C_f^{oper}/\partial P = -(\varsigma_f^*)^\top \partial b_f(P)/\partial P$ for any dual optimizer ς_f^*. Only the market-clearing rows of $b_f(P)$ depend on P, and their duals are precisely μ_{fmct}. The last equality follows by the chain rule because $\partial b_f/\partial P_{fmct}$ is zero everywhere except at the (f, m, c, t), entry, where it equals $\partial D_{fmct}/\partial P_{fmct}$. Economically, μ_{fmct} is the all-inclusive marginal system cost of serving one extra demand unit in (m, c, t), already internalizing production, transport, inventory, award, capacity, carbon and green-technology prices through the LL KKT system (Allende and Still., 2013; Kuhn and Tucker., 2013; Kjeldsen., 2000; Karush., 1939).

Sketch of proof (*3.2.1*). Differentiate Π_f in (54) with respect to P_{fmct}. The direct derivative of the revenue term is $D_{fmct} + P_{fmct}^{price} \partial D_{fmct}/\partial P_{fmct}$. The derivative of the LL value $-\mu_{fmct} D_{fmct}$ equals $-\mu_{fmct} \partial D_{fmct}/\partial P_{fmct}$ by Lemma 1. Adding both gives (56). Substituting the constant-elasticity derivative yields (57).

Sketch of proof (*3.2.1*). On any region of $\wp$ where the LL active set is fixed, μ is constant and the right-hand side of (57) is continuous and single-valued in P_{-f} ; projecting onto $\wp$ preserves nonemptiness, convexity, and upper hemicontinuity. Finite unions of such regions preserve upper hemicontinuity (Berge's maximum theorem arguments (Berge., 1984)). Continuity comes from continuity of D_{fmct} in P and piecewise constancy of μ_{fmct}, Strict concavity of Π_f in P_f on a fixed active set follows from the constant-elasticity form, since $(P - \mu)P^{-\beta}$ has a unique maximizer when $\beta > 1$. Kakutani's fixed-point theorem (Glicksberg., 1952; Kakutani., 1941) applies on the compact convex $\wp$, yielding a fixed point.

Sketch of proof (*3.2.2*). In log-price space, $\partial \Phi_{fmct}/\partial s_{fmct} = D_{fmct}(1 - \beta_{mct})$ and cross-partials $\partial \Phi_{fmct}/\partial s_{f'mct} = D_{f'mct} \eta_{mct}^{ff'}$ come from (25). Diagonal dominance (58) makes the pseudogradient strictly monotone. The dependence of μ_{fmct} on rivals' prices is weak (piecewise constant or nonincreasing under (A2)), preserving monotonicity. holds and the dependence of \mu_{fmct} on rivals' prices is insensitive (nonincreasing) (true on any fixed active-set region of the lower level), which holds piecewise under a fixed LL active set. Then the symmetric part of the Jacobian is negative definite, the pseudogradient is strictly monotone, and the price equilibrium is unique. It means Rosen's condition applies and yields uniqueness. Comparative statics follow directly from (57) discussed in Corollary 1.

Sketch of proof (*3.2.2*). Write $\Pi^{Coal}(P) = \sum_{fmct} (P_{fmct}^{price} - \mu_{mct}^c) D_{fmct}$ up to price-independent terms. Differentiate w.r.t. P_{fmct} the direct term contributes $D_{fmct} + P_{fmct}^{price} \partial D_{fmct}/\partial P_{fmct}$, the envelope term subtracts $\mu_{fmct} \partial D_{fmct}/\partial P_{fmct}$, Cross-effects add $\sum_{f \neq f'} \left(P_{f'mct}^{price} - \mu_{mct}^c\right) \partial D_{f'mct}/\partial P_{fmct}$. Substituting $\partial D_{fmct}/\partial P_{fmct} = -\beta_{mct} D_{fmct}/P_{fmct}$ and $\partial D_{f'mct}/\partial P_{fmct} = +\eta_{mct}^{ff'} D_{f'mct}/P_{fmct}$ and rearranging gives the stated markup.

Sketch of proof (*3.2.2*). In log-price space the coalition's revenue in each (m, c, t) is a strictly concave function when own-elasticity dominates the internalized cross-effects; the lower-level cost term does not depend directly on P beyond the envelope structure already differentiated out. Summing over firms preserves strict concavity, so the optimizer is unique. With the coordinated markup in Proposition 3, and as long as price bands do not bind, the comparative statics are

$$\frac{\partial P^{*}{}_{fmct}}{\partial \mu^{c}_{mct}} = \left[\beta_{mct} - \sum_{f\neq f'} \eta^{ff'}_{mct} \frac{D_{f'mct}P_{fmct}}{D_{fmct}P_{f'mct}}\right]^{-1} > 0 \quad , \quad \frac{\partial P^{*}{}_{fmct}}{\partial \ell_{fmct}} = \frac{O_{mct}}{\beta_{mct} - \sum_{f\neq f'} \eta^{ff'}_{mct} \frac{D_{f'mct}P_{fmct}}{D_{fmct}P_{f'mct}}} > 0 \qquad \text{(70-A)}$$

Therefore, sweeping or green activation that reduces μ^{c}pushes down coordinated posted prices for a given markup denominator, while better service reduces $P^{price} - P$ and hence allows lower posted P.

Sketch of proof (*3.2.2*). This is the standard capacity-investment complementary-slackness rule. The Lagrangian contribution of (67) is $\psi^{,g}_{k't}\left(\sum_{fmcp} x^{v}_{fk'mcpt} - \overline{Q}^{DC,g}_{k't}\, y^{green}_{k't}\right)$At optimum, the marginal value of one more unit of capacity is $\psi^{,g}_{k't}$. The present value of the capacity benefit is $\sum_{t} \delta_{t}\, \overline{Q}^{DC,g}_{k't}\, \psi^{,g}_{k't}$. If this exceeds the present value of the fixed activation costs, the KKT condition on $y^{green}_{k't}$prefers switching it on; if the shadow value is zero everywhere, there is no gain from activation.

Sketch of proof (*3.2.3*). For constant elasticity demand $D(P) = kP^{-\beta}$, the Hicksian area under the demand curve above the price line is$\int_{p}^{\infty} kP^{-\beta}\, dp = \frac{k}{\beta-1} P^{1-\beta}$, which equals $\frac{PD}{\beta-1}$. Summing over firms and segments gives the expression. The welfare comparison follows by algebra and the fact that coordination both lowers costs (network effects) and raises markups (price internalization).

Sketch of proof (*3.2.3*). Firm *f's* objective is $\sum_{mc}\left(P^{price}_{fmct} D_{fmct} - C^{oper}_{f}\right)$. Differentiate w.r.t$P_{fmct}$the revenue derivative is$D_{fmct} + P^{price}_{fmct}\, \partial D_{fmct}/\partial P_{fmct}$, Apply (71) for the cost term. Rearranging yields the stated First-Order Condition (FOC) and, with constant elasticity, the Lerner rule. If a price floor, cap, inter-period smoothness, or channel-parity constraint binds for some (f, m, c, t), the first-order condition (73) is replaced by the corresponding KKT complementarity condition. In this case, the Lerner-type expression describes the unconstrained interior target, which is clipped by the active bound (Lerner., 1934). This does not affect existence or uniqueness of equilibrium but only determines whether the optimal price is interior or boundary-constrained.

Sketch of proof (*3.2.3*). Under constant-elasticity demand (25)– (26), the own-price term contributes curvature proportional to β_{mct}, while cross-price effects contribute off-diagonal terms proportional to $\eta^{ff'}_{mct}$. Binding shared constraints introduce additional coupling via common multipliers in the KKT system; κ_{mct}bounds the induced cross-firm effect on the VI Jacobian. Condition $\beta_{mct} > \sum_{f'\neq f} \eta^{ff'}_{mct} + \kappa_{mct}$ yields uniform diagonal dominance of the symmetric Jacobian, implying strong monotonicity. Uniqueness then follows from standard VI theory on strongly monotone operators over convex compact sets. can be taken as any uniform bound on the magnitude of the additional off-diagonal terms in the symmetric Jacobian induced by common-dual coupling constraints; in particular $\kappa_{mct} = 0$when shared constraints are slack.

Sketch of proof (*3.2.3*). For each Y, Stage 1 has a unique equilibrium by Remark 5. The Stage 0 feasible set is a finite union of compact polyhedra; the objective is continuous in Y because $P^{*}(Y)$is continuous under a unique interior equilibrium. Weierstrass' theorem guarantees an optimizer (Stone., 1948). Backward induction gives subgame-perfectness.

Sketch of proof (*3.2.3*). In Stage 0 the firms maximize $\sum_{f\in\mathcal{F}} \Pi_{f}$over Y; choosing the "no-sharing" option $Y = Y^{NC}$reproduces the non-cooperative network, hence $\sum_{f} \Pi^{HYB}{}_{f} \geq \sum_{f} \Pi^{NC}{}_{f}$. The cooperative planner controls prices as well as Y, so it can replicate the hybrid outcome and weakly improve on it, giving the upper inequality.

Sketch of proof (*3.2.3*). Consider $\phi(Y) = \sum_{f} \Pi_{f}\, (P^{*}(Y), Y)$. By the envelope theorem for parameterized Nash equilibria, $\partial\phi/\partial Q = \sum_{f} \partial\Pi_{f}/\partial Q$because $\partial\Pi_{f}/\partial P_{f} = 0$ at P^{*}; the partial derivative equals the capacity dual at the lower level evaluated at equilibrium flows. Comparing the marginal present value $\delta_t \psi'^{g}_{k' t}$ to the average fixed-cost rate gives the activation test; if the shadow value is zero, there is no first-order gain.

4.8. Full algorithm and pseudo-code

The following pseudo-code mirrors an implementation in JuMP/Pyomo with CPLEX/Gurobi (master and LP) and PATH/IPOPT (oracles). It integrates the feasibility logic, equilibrium solves, DCEC calibration, and stabilization; it is faithful to the narrative above.

```
Algorithm : LBB-EO with Dual-Calibrated Equilibrium Cuts (DCEC)

1:  Input data (sets, costs, capacities, distances, elasticities, taxes, bands);
2:  Set termination controls: MaxIter, ε_gap, ε_LP, ε_MCP, Δ_rel ∈ [0.01,0.05], ρ ∈ [1e−4,1e−3];
3:  Choose scenario SCEN ∈ {NC, HYB, COOP};             ▷ NC/HYB = Nash MCP; COOP = centralized concave NLP
4:  Build ε-nearest sweeping sets N_j^ε and access rule C_j; prune dominated arcs;
5:  Define affine capacity map κ(Y) (prod/DC/green-DC/supplier/mode caps);
6:  Initialize Master MILP over (Y ∈ {0,1}^n, θ ∈ ℝ) with budgets, logic (38)–(40), cost-share (46); θ ≤ Θ_min;
7:  Initialize UB ← +∞, LB ← −∞, Δ_H ← Δ_0;  CutPool ← ∅;  it ← 0;

8:  while it < MaxIter and (UB − LB)/max{1,|UB|} > ε_gap do
9:     (Y^r, θ^r) ← Solve Master MILP with CutPool and trust-region ‖Y − Ŷ‖_1 ≤ Δ_H;    ▷ Ŷ = current incumbent
10:    LB ← Objective(Master);                                    ▷ global lower bound

11:    # Operations oracle (exact LP/KKT)
12:    Build LP:  minimize (1)  subject to (2),(3)–(5),(9)–(11),(14),(16)–(17);        ▷ service proxy if endogenous
13:    status_LP, (z_LP, duals_LP) ← Solve LP to ε_LP;
14:    if status_LP = infeasible then
15:       (α,β) ← Farkas certificate projected to capacity space;                  ▷ cover over κ(Y)
16:       Add feasibility cut αᵀ κ(Y) ≥ β to CutPool;  goto line 27;

17:    # Price equilibrium oracle (scenario-specific)
18:    if SCEN = COOP then
19:       Solve coalition log-price NLP in s = ln P with bands (33), smoothness (43), parity (44);
20:       Each evaluation solves Operations LP (lines 12–13); stop at KKT residual ≤ ε_LP;
21:    else  # SCEN ∈ {NC, HYB}
22:       Form MCP: price-KKT for both firms (with (33),(43),(44)) + operations-KKT; add small proximal ρ‖s−s_prev‖²/2;
23:       Solve MCP with PATH to natural residual ≤ ε_MCP;
24:    end if
25:    Extract Φ(Y^r), P*, flows z*, and capacity duals λ_raw ∈ {ξ, γ, γ^g, σ, ψ} (at equilibrium);
26:    UB ← min{UB, Φ(Y^r) − FixedActivation(Y^r)};  update incumbent (Ŷ, P̂, ẑ, λ̂) if improved;

27:    # Dual-Calibrated Equilibrium Cut (DCEC)
28:    c* ← argmax_c |λ_raw,c|;  Δ_c ← max{1, Δ_rel · κ_c(Y^r)};  κ̃ ← κ(Y^r);  κ̃_c* ← κ̃_c* + Δ_c;
29:    if SCEN = COOP then
30:       Φ_Δ ← Re-solve coalition log-price NLP at capacity κ̃;                  ▷ same tolerances
31:    else
32:       Φ_Δ ← Re-solve MCP at κ̃ with proximal weight ρ in ln-prices;              ▷ keep active set stable
33:    end if
34:    λ_cal ← λ_raw;  λ_cal,c* ← (Φ_Δ − Φ(Y^r)) / Δ_c;                          ▷ one-sided secant slope
35:    Add optimality cut  θ ≤ Φ(Y^r) + ⟨λ_cal, κ(Y) − κ(Y^r)⟩  to CutPool;
36:    Update trust-region radius Δ_H (expand if last cut strongly violated; shrink on oscillations);

37:    it ← it + 1;
38:  end while

39:  Return Y*, P*, z*, duals* from incumbent; also report:
40:     • Master gap (UB − LB)/max{1,|UB|} ≤ ε_gap,
41:     • LP primal/dual infeasibilities ≤ ε_LP, complementarity sum ≤ ε_LP,
42:     • MCP natural residual (NC/HYB) ≤ ε_MCP (or NLP KKT residual in COOP).
```

Figure 2. Pseudo-code of our proposed LBB–EO–DCEC.

What the pseudo-code is doing in mathematical terms. Each master iteration proposes a design Y^r and a lower bound. The LP oracle either returns a Farkas feasibility certificate, which becomes a linear cover inequality over $\kappa(Y)$, or it yields primal flows and duals. The price oracle returns an equilibrium consistent with (NC/HYB) MCP or COOP concave log-price program. The algorithm then forms an optimality cut that is linear in Y because capacities are affine in Y. When prices are at corners, the DCEC replaces the raw dual slope by a secant computed from two exact equilibrium values, ensuring validity and improving tightness. Because the design space is finite and each revisit yields a new feasibility or optimality cut, the loop terminates finitely once the upper–lower gap is within tolerance. The returned solution is a structure Y^* with true equilibrium prices P^* and flows that satisfy all constraints, together with certificates (LP residuals, MCP residual, master gap) that reviewers can use to verify optimality and feasibility.

4.9. Relationship to Standard Logic-Based Benders Decomposition (LBBD)

The proposed solution framework builds on the logic-based Benders decomposition (LBBD) paradigm but departs from the classical formulation in several fundamental and necessary ways to accommodate strategic pricing equilibria, non-smooth service effects, and carbon-regulated coopetition.

In standard LBBD, the original problem is decomposed into (i) a master problem that optimizes over discrete design or policy decisions and (ii) one or more continuous subproblems, typically linear or convex nonlinear programs. Given a fixed master design, the subproblem either returns an optimal objective value together with valid dual multipliers used to generate Benders optimality cuts or produces an infeasibility certificate (e.g., a Farkas ray) that yields feasibility cuts. This classical structure relies critically on two assumptions: (i) the subproblem is a well-defined convex optimization problem, and (ii) its dual multipliers provide globally valid marginal values for constructing supporting cuts.

While our framework shares the high-level master–subproblem separation of LBBD, these assumptions do not hold in the present setting. Here, feasibility and optimality are not determined by the operational LP alone. Instead, the realized objective value depends on an equilibrium-consistent pricing layer, which fundamentally alters the nature of the "subproblem." Specifically, conditional on a candidate strategic design Y^r, the lower layer consists of two tightly coupled components: (i) an exact operational LP that determines production, inventory, multimodal logistics, service levels, and emissions, and (ii) a price equilibrium problem that governs market outcomes. In the NC and HYB regimes, prices arise from a generalized Nash equilibrium formulated as a monotone MCP. In the COOP regime, prices result from a centralized but constrained concave pricing optimization. Consequently, the "subproblem" is not a single convex program but an equilibrium oracle whose solution must simultaneously satisfy operational feasibility and equilibrium optimality conditions.

This distinction is crucial. Although LP dual multipliers correctly price marginal changes in capacities and flows, they may misrepresent the true marginal impact on the equilibrium objective when prices are restricted by policy bands, parity constraints, inter-temporal smoothness limits, or when service levels are governed by non-smooth operators such as mi n·or saturation functions. Near such kinks, classical Benders optimality cuts constructed from raw duals tend to be weak or misleading, allowing the master to propose designs that appear profitable in relaxation but are not attainable under equilibrium behavior. As a result, a direct application of standard LBBD may converge slowly or oscillate. To address these limitations, we extend the LBBD framework along two key methodological dimensions.

First, we replace conventional subproblem solvers with Equilibrium Oracles (EO).

Given a candidate design Y^r, the oracle computes a fully consistent equilibrium outcome prices, flows, service levels, and emissions by solving either (i) a Nash MCP (NC/HYB) or (ii) a concave coalition pricing problem (COOP), embedding the exact operational LP. Feasibility is therefore defined not only by operational constraints but also by equilibrium conditions. This guarantees that every incumbent solution, reported bound, and oracle evaluation is economically meaningful and equilibrium-consistent.

Second, we introduce Dual-Calibrated Equilibrium Cuts (DCEC).

Rather than relying solely on raw dual multipliers from the operational LP, DCEC locally calibrate marginal values using one-sided secant slopes of the equilibrium value function with respect to selected components of the affine capacity map $\kappa(Y)$. These slopes are computed by applying a small, equilibrium-preserving perturbation to the design, re-solving the equilibrium oracle, and measuring the resulting change in the equilibrium objective. Because DCEC are constructed from exact oracle evaluations, they remain valid lower-bounding cuts while being substantially tighter than classical Benders tangents in the presence of non-smooth price or service effects.

From a methodological standpoint, the proposed LBB–EO–DCEC framework can therefore be viewed as a generalization of logic-based Benders decomposition to equilibrium-constrained design problems. Compared with standard LBBD, it (i) replaces convex subproblems with equilibrium oracles, (ii) augments dual-based cuts with calibrated equilibrium slopes, and (iii) preserves a pure MILP master with certified upper and lower bounds. Compared with monolithic MPCC or EPEC reformulations, it avoids large nonconvex mixed-integer programs and instead achieves numerically stable convergence through decomposition, with explicit feasibility and optimality certificates.

In summary, the proposed approach retains the conceptual simplicity of LBBD a master–oracle architecture with iterative cut generation while extending it in a principled way to handle strategic pricing, non-smooth service effects, and carbon-regulated coopetition that a standard decomposition cannot capture. This positioning clarifies both the rigor and the practical value of the method for large-scale, equilibrium-driven supply chain design problems.

Appendix A. Literature Synthesis and Comparative Positioning

Table A1. Tabular summary of the literature (pricing–inventory–carbon–marketing–games)

Study	Supply chain setting	Core decisions	Carbon / policy instrument	Demand / behavior features	Modeling / solution approach	Main takeaways (managerial + theoretical)
Choudhury et al. (2025)	Retailer-centric (multi-product categories; supplier offers volume discount + trade terms)	Retail price, advertising, service investment, cycle time	Cap-and-price, cap-and-trade, carbon tax	Demand influenced by price + promotion + service	Analytical structure + novel algorithm, numerical + sensitivity	Jointly optimizing marketing + operational levers can improve profit and emissions performance; regulation choice matters—policy regimes change the "best" operating region and investment intensity.
Sun et al. (2025)	1 supplier – 1 manufacturer	Low-carbon R&D (abatement level), strategic mode choice	Carbon tax (SDG framing)	Market/environment performance metrics (profit, consumer surplus, emissions)	Equilibrium comparisons across cooperative vs independent R&D modes	A sufficiently high tax can stimulate joint abatement when R&D cost coefficient is low; "simultaneous independent R&D" can dominate single-firm R&D in abatement and welfare; cooperation only-in-R&D can yield maximal abatement in their extension.
Yan & Ni (2024)	Manufacturer–retailer	Selling prices + emission reduction rates	Cap-and-trade	Uncertainty theory: parameters as uncertain variables (expert belief)	Decentralized game models; analytical solutions + numerical	Under limited data and non-frequency-stable parameters, uncertain-game modeling produces tractable equilibria and useful comparative insights across decentralized structures.
Yadav et al. (2024)	Deteriorating inventory system (single firm / retailer view)	Selling price, preservation technology, carbon reduction investment	Emission reduction measures (regulatory context)	Price-dependent demand; deterioration slowed by PT; waste/salvage	Inventory model + numerical/sensitivity	Preservation investment tied to price can be profit-improving and emission-reducing; highlights role of PT + abatement as joint levers, with salvage handling.
Mahato et al. (2024)	Sustainable inventory with controllable deterioration + trade credit	Price, preservation investment, cycle time	Multiple regimes: cap-and-price, carbon tax, cap-and-trade, none	Deterioration controllable; financing via trade credit	Theoretical properties + algorithm + examples/sensitivity	Carbon policy ranking can differ: in their comparisons, cap-and-price can yield higher profit than other regimes; trade credit interacts with sustainability levers and deterioration control.
Zhang & Yu (2023)	Low-carbon closed-loop SC: manufacturer–retailer–government	Pricing, emission-reduction investment, recycling investment; government subsidies	Government double subsidy + dynamic setting	Goodwill and recycling rate dynamics	Three-level differential games; equilibrium derivations + sensitivity	Power structure shapes subsidy rates and private investments; weaker party tends to receive higher subsidy; non-led structure may improve recycling/social welfare under subsidies; contracts can coordinate.

Li & Mizuno (2022)	Dual-channel manufacturer–retailer	Dynamic pricing + inventory control	None (baseline RM / DP)	Stochastic price-sensitive demand	Stochastic dynamic programming; power structure comparisons	Optimal policy structure resembles base-stock + list-price type; power structure changes levels (not structural form); leadership affects distribution of benefits and decisions.
Pourmohammad-Zia et al. (2021)	Food SC with growing + deteriorating items	Dynamic pricing + inventory; coordination contract	None	Growth function + deterioration; waste considerations	Nonlinear convex optimization; centralized vs decentralized + profit-sharing	Coordination can reduce prices and increase total profit; shows how biology/growth + deterioration reshape dynamic pricing and replenishment timing.
Sheng et al. (2023)	Pricing strategy selection	Dynamic pricing vs pre-announced pricing	None	Informed/uninformed + patience heterogeneity	2-period game-theoretic analysis	Which pricing regime dominates depends on consumer mix and patience; decentralized setting can still yield “win-win-win” regions for firms + consumers.
Ranjan & Jha (2022)	2-echelon deteriorating product SC	Multi-period dynamic retail pricing; replenishment cycle; PT investment	Carbon emission included	Demand depends on price, inventory, reference price, freshness	Algorithmic optimization + numerical/sensitivity	PTI + dynamic pricing can strongly raise profits; reference price is a key driver; carbon costs shift the PTI–pricing trade-off frontier.
Parilina et al. (2025)	Manufacturer–retailer (Stackelberg differential game)	Dynamic pricing + advertising (and contract choice)	“Environment- and fairness-concerned consumers”	Reference price fairness + green info sensitivity	Differential game; compare wholesale vs revenue-sharing	Contract form changes whether outcomes can be Pareto-improving (members + consumers + environment); fairness/green disclosure reshapes optimal pricing/advertising paths.
Zarouri et al. (2022)	Perishable product SC	Two-period markdown pricing; cooperation vs Stackelberg	None	Stochastic, quality-dependent demand	Heuristic analysis of games	Cooperation yields lowest prices and highest total profit; leadership changes price markdown intensity; highlights coordination value under perishability.
Yan et al. (2024)	Manufacturer + digital retailer + B&M retailer	Dynamic pricing + emission reduction + promotions	Nonexplicit (green reputation)	Bidirectional free-riding, reputation dynamics	Tripartite dynamic game	Free-riding fundamentally changes how promotions and reputation translate into prices/profit; reputation effects can flip depending on cross-effects; parameters (decay, cost coefficients) define regimes.
Rahmani & Pashapour (2024)	Dual channel with new + returned products	Prices for new/refurbished + network decisions (bi-objective MINLP)	Sustainability framing	Customer segmentation: return behavior modeled	Bi-objective MINLP + metaheuristics	More behavioral return modeling improves realism; dynamic prices across segments can improve profit while meeting social objective.

Shu et al. (2025)	Two-period CLSC with collection inconvenience	Pricing + collection investment	Nonexplicit	Channel inconvenience; competition in collection	Game-theoretic + numerical	Collection competition improves environmental performance but hurts profits and welfare; coordination contract (two-part tariff) can restore performance.
Alamdar & Seifi (2024)	Retailer with strategic customers; seasonal products	Ordering + dynamic pricing	None	Strategic waiting; multiple substitutes	Deep Q-learning (RL) + real dataset	RL can outperform classic metaheuristics in large state spaces; demand learning + pricing/ordering integration is powerful when closed-form optimization is hard.
Bai et al. (2024)	Advance selling policies	Pricing + ordering under advance selling variants + advertising	None	Stochastic valuations: advertising affects diffusion	Analytical solutions + numerical	The best policy depends on share of high-valuation customers; advertising changes diffusion and profitability; identifies thresholds for when advance selling dominates.
Zhou et al. (2022)	On-site services (routing + pricing)	Vehicle routing (CO_2) + time-window pricing	Emission reduction objective	Customers choose time window based on price	Two-phase: low-emission VRP then pricing optimization; metaheuristic for online use	Dynamic pricing can shift demand to cleaner schedules and raise profit; useful template for coupling “emissions-minimizing ops” with “profit-maximizing pricing.”
Schlosser et al. (2021)	Circular economy (firm-level control)	Dynamic pricing + recycling investment	Circularity objective (recycling rate)	Recycling affects supply + demand, diminishing effect over time	Optimal control; steady state + sensitivity	Joint control of price and recycling can stabilize sustainability trajectories; investment combats the natural decay of “green demand response.”
Tunçinan et al. (2024)	1 warehouse – many heterogeneous retailers	Allocation + pricing over horizon	None	Deterministic price-sensitive demand; lost sales	MIQCP + structural results + heuristic	When inventory is large, tractability improves; retailer heterogeneity & seasonality drive optimal price/allocation paths; decomposition heuristic yields small gaps.
Soares et al. (2020)	Electricity retail tariffs	Bi-level TOU pricing	None	Consumer response in lower level; lower-level hard to solve exactly	Bi-level + bound estimation under computation budget	When LL can’t be solved exactly, bounding becomes essential for reliable leader decisions; a strong methodological reference for BL settings.
Gholami et al. (2021)	Bi-level channel optimization, multi-period	Dynamic pricing with stochastic demand memory	None	Price-dependent demand with memory	Iterative algorithm that decouples nested equilibria	Demand “memory” makes equilibria nested; iterative decoupling gives explicit equilibria across broad scenarios/time horizons.
Song et al. (2021)	Seller–buyer dynamic Stackelberg	Wholesale price + innovation efforts over time	None	Goodwill dynamics: innovation affects demand and cost	Differential Stackelberg game	Innovation + pricing coevolve; leadership changes profit and goodwill paths; useful for “dynamic effort + price” modeling.

Pazoki et al. (2024)	Regulated transport capacity (Canadian case)	Equilibrium pricing + quota regulation	Government intervention (quota)	Disruption-driven capacity scarcity	Market models + algorithm	Justifies when intervention is necessary; quota stabilizes outcomes under capacity scarcity; bridges resilience + regulation + equilibrium pricing.
Wang & Du (2025)	Incentive programs	Price discrimination + quality discrimination	Data compliance policy framing	Customer incentive programs + patience	Analytical comparisons	Firms may offer lower prices + higher quality to attract new customers; rewarding old customers can be suboptimal; policy constraints affect welfare.
Dye (2020)	Perishable items; multi-period	Dynamic pricing + advertising + inventory (psychic stock)	None	Demand depends on price, freshness, goodwill, display stock	Infinite-horizon dynamic optimization	Under some settings, optimal pricing can become static while goodwill/stock dynamics converge; gives structural properties for DP-advertising-inventory coupling.
Qiu et al. (2023)	Retailer with quick response	2-period pricing + ordering	None	Distributionally robust market size uncertainty	DRO reformulations; closed-form conditions	Dynamic pricing benefits depend on uncertainty and quick response; DRO gives tractable robustness when distribution is unknown.
Zhou et al. (2023)	Perishable, partial lost sales	Pricing + two-stage inventory (reorder/return)	None	Random demand; old vs fresh competition	Dynamic programming	Mid-season reorder decreases with leftover quantity/quality; shows how "quality-differentiated inventory" shapes pricing and replenishment.
Dye & Hsieh (2024)	Green innovation + pricing + advertising	Dynamic pricing + green innovation + advertising	Cap-and-price	Demand multiplicative in price and goodwill; goodwill evolves	Infinite-horizon dynamic optimization + structural results	Green innovation and advertising build goodwill; cap-and-price changes steady-state and policy sensitivities; provides interpretable structural effects.
Crettez et al. (2025)	Competing supply chains with coordination	Capacity choice + retail price under revenue-sharing	None	Complementary goods; discontinuity issues	Differential game; equilibrium existence analysis	Revenue-sharing can destroy equilibrium existence due to discontinuities—important caution for "coordination contracts + competition" modeling.
Hamidoğlu & Weber (2024)	Low-carbon agriculture; collaborative market	Low-carbon adoption efforts + collaborative pricing	Subsidies + carbon taxes	Cooperative Nash interactions	Nash-based operational framework + case studies	Proper subsidy/tax design stabilizes low-carbon transition; collaboration can align adoption and affordability; policy can reduce cost gap vs conventional.
Han et al. (2025)	Dyadic SC with fairness concerns	Pricing under power structures	None	Fairness concerns affect decisions	Stackelberg vs Nash comparisons	Fairness of leaders increases retail prices; power balance changes profit distribution and total performance; fairness expands pricing theory under power structures.

Table A2. Comparative positioning of this study vs. state of the art

Dimension	Representative state-of-the-art studies	What existing studies do	Key limitation in literature	This study (MEMM-SCN)
Supply chain scope	Li & Mizuno (2022); Sun et al. (2025); Yan & Ni (2024)	Focus on single- or two-echelon supply chains; usually one market	Cannot capture cross-echelon and cross-market spillovers	Multi-echelon, multi-market, multi-channel network with endogenous flows
Pricing structure	Li & Mizuno (2022); Sheng et al. (2023); Wang et al. (2024)	Dynamic pricing, typically single-channel or dual-channel	Pricing is not linked to network topology or logistics decisions	Channel-specific dynamic pricing jointly optimized with network configuration
Carbon regulation modeling	Choudhury et al. (2025); Sun et al. (2025); Yan & Ni (2024); Dye & Hsieh (2024)	Carbon tax, cap-and-trade, or cap-and-price studied in isolation	Carbon policy effects analyzed without endogenous logistics pooling	Multiple carbon regimes embedded directly in firm-level constraints
Sustainability levers	Yadav et al. (2024); Ranjan & Jha (2022); Schlosser et al. (2021)	Investment in abatement, preservation, or recycling	Sustainability decisions treated separately from pricing competition	Carbon mitigation + service + pricing decisions co-optimized
Inter-firm behavior	Sun et al. (2025); Zhang & Yu (2023); Shu et al. (2025)	Either pure competition or full cooperation	Binary behavioral assumption unrealistic in practice	Three regimes: NC, COOP, and novel HYB coopetition
Nature of cooperation	Zhang & Yu (2023); Yi et al. (2024)	Cooperation via contracts or centralized control	No partial cooperation preserving price competition	Shared logistics & green assets + price competition (coopetition)
Game-theoretic structure	Li & Mizuno (2022); Zhang et al. (2025 IEEE TNSE)	Nash or Stackelberg games, usually separable	Cannot handle shared constraints across firms	Generalized Nash Equilibrium (GNE) with shared logistics and carbon constraints
Optimization hierarchy	Soares et al. (2020); Gholami et al. (2021)	Bi-level models focused on pricing or tariffs	Network design rarely appears at strategic level	Bi-level: strategic GNE (upper) + operational profit maximization (lower)
Mathematical complexity	Tunçinan et al. (2024); Rahmani & Pashapour (2024)	MILP / MINLP without equilibrium consistency	Discrete network + equilibrium often approximated	Multi-level MINLP with equilibrium constraints
Solution methodology	Classical Benders; heuristics; metaheuristics (Alamdar & Seifi, 2024)	Heuristics lack equilibrium guarantees	No equilibrium-consistent decomposition	Logic-Based Benders with Equilibrium Oracles (LBB-EO) + DCEC
Logistics–pricing interaction	Zhou et al. (2022); Pazoki et al. (2024)	Pricing affects routing or quotas exogenously	No endogenous topology reconfiguration	Pricing reshapes topology and modal split (truck → rail)
Resilience under constraints	Pazoki et al. (2024); Zhang & Yu (2023)	Focus on single disruption or policy stress	Strategic regime robustness unexplored	COOP shown most robust under rail scarcity & binding quotas

Empirical grounding	Many studies rely on stylized or synthetic data	Limited industrial realism	Results are hard to operationalize	Real electronics supply chain case with empirical demand, carbon prices
Managerial insight level	Choudhury et al. (2025); Mahato et al. (2024)	Pricing- or inventory-centric insights	Cannot explain structural shifts	End-to-end insights: topology, emissions, profitability, resilience

Appendix B. KKT Implementation of the Lower-Level Model and Single-Level Reformulations

B.1. *Primal LP in standard form (given strategic binaries and prices)*

Fix a scenario (NC/COOP/HYB), a strategic activation/investment plan (plants, DCs, green DCs, awards, sweeping assignments), and a posted price vector P. The lower level minimizes augmented operating cost (production, transport, inventory, and any carbon tax/permit cost pushed into unit costs per section 3.1), subject to feasibility. Stack all nonnegative continuous lower-level variables in z (these include all flows x, production q, inventories I, and if the service proxy is embedded auxiliaries N^v, T, S^v, s see section A.5). Write the LP as

$$min_{Z\geq 0} \mathcal{C}^\top z \qquad \text{s.t.} \qquad \mathrm{A}z \ = b(P,\ell)\,, \qquad Gz \leq h \tag{KKT.1}$$

Where $A\mathcal{Z} \ = b$, collects equalities (plant outbound balance (16), DC and green-DC inventory dynamics (9)–(10), market clearing (11), definitions for N^v, T, S^v, sif used), and $Gz \leq h$ collects capacities and gating ((2)–(5), (14), (17), sweeping and award linkages from (59)–(61) and (31)–(32)). The right-hand side $b(P,\ell)$ contains the demand $D_{fmct}(P,\ell)$ in (11). if ℓ is endogenized we add its defining constraints ((19)–(23)) to $A\mathcal{Z} \ = b$ as in section A.5. Unit-cost vector c is already augmented for carbon per section 3.1.2 (so the KKT remains linear). Indexing and arc categories used below.

Supplier→plant x^v_{ifjat}; plant→private-DC x^v_{fjkpt}; plant→green-DC $x^v_{fjk'pt}$; private-DC→market x^v_{fkmcpt}; green-DC→market $x^v_{fk'mcpt}$. Production q^{prod}_{fjpt}; private-DC inventory I_{fkpt}; green-DC inventory $I_{fk'pt}$. Sweeping assignment binaries $w_{ijat}$and award binaries $y^{\text{sup}}_{iat}$are upper level (fixed here).

B.2. *Dual, KKT, and economic multipliers*

Let y be the free dual vector for equalities $Az = b$ and $u \geq 0$the multipliers for capacities $Gz \leq h$. The dual is

$$min_{y^{free}, u\geq 0} b(P,\ell)^\top y + h^\top u \qquad \text{s.t} \qquad \mathrm{A}^\top y + G^\top u \ \leq c \tag{KKT.2}$$

The KKT system is:

- *Primal feasibility:* $Az = b(P,\ell)$, $Gz \leq h$, $z \geq 0$.
- *Dual feasibility:* $A^\top y + G^\top u \leq c$, $u \geq 0$.
- *Complementary slackness (pairwise):* for each capacity row g, $u_g \ \perp \ (h_g - g^\top z)$; for each nonnegative variable z_i, $z_i \ \perp \ r_i$, where the reduced cost

$$r = c - \mathrm{A}^\top y + G^\top u \ \textit{satisfies}\ r \geq 0 \tag{KKT.3}$$

We name key duals for interpretation:

- ϕ_{fjpt}(free): plant outbound balance (16).
- $\eta_{fkpt}$and $\eta^g_{fk'pt}$(free): private and green DC inventory dynamics (9)– (10).
- μ_{fmct}(free): market clearing (11) (this is the "all-inclusive marginal cost" used in price FOCs).
- $\xi_{fjpt} \geq 0$: production capacity (2).
- $\gamma^{\text{in}}_{fkpt}, \gamma^{\text{out}}_{fkpt} \geq 0$: private-DC inbound/outbound (4)– (3).
- $\gamma^g_{k't} \geq 0$: green-DC shared capacity (5).
- $\sigma_{iat} \geq 0$: supplier capacity (14).
- $\psi_{vt} \geq 0$: mode capacity (17).
- $\pi_{iat} \geq 0$: award-gate (sum over f, j, vflows $\leq \bar{M}_{at} y^{\text{sup}}_{iat}$).

- $\varpi_{ijat} \geq 0$: sweeping-gate $x_{ifjat}^{v} \leq \bar{M}_{jat} w_{ijat}$.

(If emissions are not pushed into c, include an emission accounting equality with dual κ_t; see section A.6.)

B.3. *Stationarity (reduced-cost) equations by variable family*

The reduced-cost form $r = c - A^{\top} y - G^{\top} u$makes signs unambiguous. For any variable $z_i > 0$, its reduced cost $r_i = 0$; otherwise $r_i \geq 0$.

Flows from supplier to plant x_{ifjat}^{v}, This variable appears in, plant material balance (6) on the left with +1 coefficient (we write it as an equality by adding a nonnegative slack for over-supply), supplier capacity (14) with +1, mode capacity (17) with +1, sweeping gate with +1, and award gate with +1. It does not enter DC or market equalities directly. Its unit cost is $c_{v,i\to j,t}^{\mathrm{Tran}}$(already carbon-augmented). Hence

$$r_{i,f,j,a,t}^{sup\to pl} = c_{v,i\to j,t}^{\mathrm{Tran}} - \alpha_{fjat} - \sigma_{iat} - \psi_{vt} - \varpi_{ijat} - \pi_{iat} \geq 0, \qquad x_{ifjat}^{v} \perp r_{i,f,j,a,t}^{sup\to pl} \tag{KKT.4}$$

where $\alpha_{fjat}$is the free dual of the material balance equality $\sum_{i,v} x_{ifjat}^{v} - \sum_{p} r_{ap} q_{fjpt}^{\mathrm{prod}} - s_{fjat}^{\mathrm{mat}} = 0$(we introduce $s_{fjat}^{\mathrm{mat}} \geq 0$to convert (6) into an equality).

Production q_{fjpt}^{prod}, This variable appears in, plant outbound balance (16) with −1, production capacity (2) with +1, material balance equalities with −r_{a p}, and (optionally) emission accounting with $+e_{fjp}^{\mathrm{prod}}$ if not pushed into cost. Unit cost is c_{fjpt}^{prod}(carbon-augmented). Then

$$r_{fpjt}^{prod} = c_{fpjt}^{prod} + \xi_{fpjt} - \sum_{p} r_{ap}\alpha_{fjat} - \underbrace{\kappa_t e_{fjp}^{\mathrm{prod}}}_{\text{Omit if embedded}} + \cdots + \left(+\emptyset_{fpjt}\right), \tag{KKT.5}$$

but careful with the sign of ϕ: since (16) is $\sum_{k,v} x_{fjkpt}^{v} + \sum_{k',v} x_{fjk'pt}^{v} - q_{fjpt}^{\mathrm{prod}} = 0$, the coefficient on $q$is −1, so $r = c - A^{\top} y - \cdots$contributes −(−1)\phi=+\phi. Hence

$$r_{fpjt}^{prod} \geq 0, \qquad q_{fjpt}^{\mathrm{prod}} \perp r_{fpjt}^{prod} \tag{KKT.6}$$

Flows from plant to private DC x_{fjkpt}^{v}, $\mathrm{This variable appears in, plant outbound balance (16) with + 1, private - DC inventory equality (9) with + 1} \left(\mathrm{we write inventory as} (1-\phi_k) I_{t-1} + \sum_{j,v} x_{fjkpt}^{v} - \sum_{m,c,v} x_{fkmcpt}^{v} - I_{fkpt} = 0\right)$, $inbound DC capacity (4) with + 1, and mode capacity (17) with + 1. Unit cost c_{v,j\to k,t}^{Tran} (carbon - augmented). Then$

$$r_{fjkpt}^{pl\to pdc} = c_{v,j\to k,t}^{Tran} - \phi_{fjpt} - \eta_{fkpt} - \gamma_{fkpt}^{in} - \psi_{vt} \geq 0, \qquad x_{fjkpt}^{v} \perp r_{fjkpt}^{pl\to pdc} \tag{KKT.7}$$

Flows from plant to green DC $x_{fjk'pt}^{v}$, Analogous to private DC x_{fjkpt}^{v} but using the green-DC inventory dual $\eta_{fk'pt}^{g}$:

$$r_{fjk'pt}^{pl\to gdc} = c_{v,j\to k',t}^{Tran} - \phi_{fjpt} - \ \eta_{fk'pt}^{g} - \psi_{vt} \geq 0, \qquad x_{fjk'pt}^{v} \perp r_{fjk'pt}^{pl\to gdc} \tag{KKT.8}$$

Flows from private DC to market x_{fkmcpt}^{v}, This variable enters, private-DC inventory equality (9) with −1, market-clearing equality (11) with +1, outbound DC capacity (3) with +1, mode capacity (17) with +1. Unit cost $c_{v,k\to(m,c),t}^{\mathrm{Tran}}$. Hence

$$r_{fkmcpt}^{pdc\to mkt} = c_{v,k\to(m,c),t}^{Tran} + \eta_{fkpt} - \mu_{fmct} - \ - \gamma_{fkpt}^{out} - \psi_{vt} \geq 0, \qquad x_{fkmcpt}^{v} \perp r_{fkmcpt}^{pdc\to mkt} \tag{KKT.9}$$

Flows from green DC to market $x^{v}_{fk'mcpt}$, Use green-DC inventory dual η^{g} and shared capacity dual γ^{g}:

$$r^{gdc\to mkt}_{fk'mcpt} = c^{Tran}_{v,k'\to(m,c),t} + \eta^{g}_{fk'pt} - \mu_{fmct} - -\gamma^{g}_{k't} - \psi_{vt} \geq 0, \qquad x^{v}_{fk'mcpt} \perp r^{gdc\to mkt}_{fk'mcpt} \qquad \text{(KKT.10)}$$

Inventories I_{fkpt} and $I_{fk'pt}$, For private DCs, the inventory equality at t is $(1-\phi_k)I_{fkp,t-1} + \text{inflow}_t - \text{outflow}_t - I_{fkpt} = 0$ with dual η_{fkpt}. I_{fkpt} also appears in period $t+1$ with coefficient $+(1-\phi_k)$. With unit holding cost c^{Inv}_{fkpt},

$$r^{Inv}_{fkpt} = c^{Inv}_{fkpt} - \eta_{fkpt} - (1-\phi_k)\eta_{fkp,t+1} \geq 0, \qquad I_{fkpt} \perp r^{Inv}_{fkpt} \qquad \text{(KKT.11)}$$

with the terminal condition $\eta_{fkp,T+1} \equiv 0$. The green-DC case is identical with η^{g} and $c^{\text{Inv}}_{fk'pt}$.

Material-balance slack s^{mat}_{fjat}, Arises when we convert (6) to equality. Since it enters with coefficient -1 and has zero-unit cost,

$$r^{mat-slack}_{fjat} = 0 - \left(-\alpha_{fjat}\right) = +\alpha_{fjat} \geq 0, \qquad s^{\text{mat}}_{fjat} \perp \alpha_{fjat} \qquad \text{(KKT.12)}$$

At optimum we typically get $s^{\text{mat}} = 0$ and $\alpha_{fjat} \leq 0$ (non-binding material balance), but writing the equality keeps the KKT system clean.

B.4. *Complementary-slackness pairs for capacities*

For every inequality row "≤", include the orthogonality $u_g \perp \left(h_g - g^{\top}z\right)$. Explicitly:

- *Production capacity* (2): $\xi_{fjpt} \perp \left(\bar{Q}^{\text{prod}}_{fjpt} + U^{\text{cap}}_{fjp}\Delta\bar{Q}^{\text{prod}}_{fjpt}\right)y^{\text{PL}}_{fjpt} - q^{\text{prod}}_{fjpt}$.
- *Private-DC inbound/outbound (4)/(3):* $\gamma^{\text{in}}_{fkpt} \perp \bar{Q}^{\text{DC}}_{fkpt}y^{\text{DC}}_{fkt} - \sum_{j,v} x^{v}_{fjkpt}$; and $\gamma^{\text{out}}_{fkpt} \perp \bar{Q}^{\text{DC}}_{fkpt}y^{\text{DC}}_{fkt} - \sum_{m,c,v} x^{v}_{fkmcpt}$.
- *Green-DC shared* (5): $\gamma^{g}_{k't} \perp \bar{Q}^{\text{DC},g}_{k't}y^{\text{green}}_{k't} - \sum_{f,m,c,p,v} x^{v}_{fk'mcpt}$.
- *Supplier capacity* (14): $\sigma_{iat} \perp S^{\max}_{iat} - \sum_{f,j,v} x^{v}_{ifjat}$.
- *Mode capacity* (17): $\psi_{vt} \perp \text{Cap}^{\text{tran}}_{vt} - \sum_{\text{all arcs on }(v,t)} x^{v}$.
- *Award gate:* $\pi_{iat} \perp \bar{M}_{at}y^{\text{sup}}_{iat} - \sum_{f,j,v} x^{v}_{ifjat}$.
- *Sweeping gate*: $\varpi_{ijat} \perp \bar{M}_{jat}w_{ijat} - \sum_{f,v} x^{v}_{ifjat}$.

(Activation binaries $y^{\text{PL}}, y^{\text{DC}}, y^{\text{green}}$ and awards y^{sup} are fixed in the lower level; their KKT appear in the upper-level MPEC if they are decision variables.)

B.5. *Service proxy inside the KKT*

If the service proxy ℓ in the demand $D(P,\ell)$ is endogenized (rather than treated as a parametric input from the upper level), include the linearized definitions (19)–(23) via auxiliary variables N^{v}_{fmct}, T_{fmct}, S^{v}_{fmct}, s_{fmct} and linearization constraints:

- Flow aggregations (21)– (22) become equalities in $Az = b$.

- Shares $S^v = N^v/T$and $s = \min 1, T/(C + \epsilon)$are linearized as in the approach (SOS2 or McCormick) and enter as additional equalities/inequalities with their own duals. Their stationarity conditions follow the same reduced-cost template and need not be written out term-by-term; solvers handle them once the linearization is explicit. Linearizing $S^v = N^v/T$(exact MILP with McCormick+SOS2 or convex relaxation)

Introduce $S^v \in [0,1]$, $T \in [0, \bar{T}]$and the identity $N^v = S^v T$(bilinear). Options:

- *Piecewise linear (SOS2) exact*: segment $T \in [0, \bar{T}]$at breakpoints $0 = \tau_0 < \cdots < \tau_K = \bar{T}$. Use convex combinations $T = \sum_k \lambda_k \tau_k$, $\sum_k \lambda_k = 1$, SOS2 on λ. Enforce $N^v = \sum_k \lambda_k \tau_k S^v$. This is linear with SOS2 and is exact on the grid.
- *McCormick envelopes (convex):* with known bounds $0 \le S^v \le 1$, $0 \le T \le \bar{T}$, add four linear inequalities for each bilinear product $N^v = S^v T$. This yields a convex outer approximation; refine with cuts or more breakpoints if needed.

- Add $T = \sum_v N^v$and $\sum_v S^v \le 1$(the latter is implied when $T > 0$; include $S^v \le 1$and $T \ge 0$). Linearizing $\min 1, u$with $u = T/(C + \epsilon)$, Let $C^+ = C + \epsilon$(constant once activations are fixed). Introduce $0 \le s \le 1$and enforce $s \le \frac{T}{C^+}, s \le 1$.
- *Piecewise linear epigraph:* segment $T \in [0, \bar{T}]$, use SOS2 to enforce $s \le T/C^+$on each segment. If exactness is desired, impose equality $s = \min 1, T/C^+$via two linear inequalities $s \le 1$and $s \le T/C^+$together with a convex, increasing penalty in the objective (or binary "kink" selection with big-M for strict equality). In practice, $s = \min 1, T/C^+$with the two inequalities is tight once ℓenters the objective or constraints. Finally,$\ell = \alpha_0 + \alpha_1(1 - s) + \alpha_2 \sum_v \theta_v S^v$ is linear once $S^v$and $s$are linearized.

- The effective price $P_{fmct}^{\text{price}} = P_{fmct} + O_{mct}\left(1 - \ell_{fmct}\right)$then appears only in the RHS of market clearing $b(P, \ell)$, preserving the KKT envelope used in the price FOCs. Parity and smoothness constraints (43)–(45) are linear inequalities. Price bounds (33) linear.

This preserves linearity (SOS2) or convexity (McCormick) of the lower level, so the KKT system remains valid (LP/MILP case) or can be written as an MCP (convex case).

B.6. *Two equivalent treatments of carbon in KKT*

Multiply each arc/production unit by its emission factor and price it at τ_t(or the endogenous permit shadow, if pooled). These terms sit in c, so no extra dual is needed, and all stationarity equations above already include carbon. Alternatively, keep an equality for emissions $E_{ft} - \sum(\text{factors} \times \text{flows}) = 0$with dual κ_{ft}(NC/HYB) or pooled $E_t^{\text{coal}}$with dual κ_t(COOP). Then each stationarity picks up a term $-\kappa \cdot$ factor, and $\kappa$is the endogenous carbon shadow price. If cap/trade constraints are also present at the lower level, add their complementary-slackness pairs (period caps, banking recursion) with the usual free/positive duals. Because demand enters only on the RHS of market clearing, Danskin's theorem gives

$$\frac{\partial C_f^{oper}}{\partial P_{fmct}} = \sum_{\rho(11)} y_\rho \frac{\partial D_{fmct}(P, \ell)}{\partial P_{fmct}} = \mu_{fmct} \frac{\partial D_{fmct}}{\partial P_{fmct}} \tag{KKT.13}$$

with the sign convention in the profit objective yielding the minus sign used in section 3.2.1–section 3.2.3. This identity is exactly what produces the NC/HYB Lerner rules and the COOP coordinated markup once we differentiate the upper-level objectives.

Single-level reformulations (MPCC/MPEC/EPEC)

- **NC:** Replace the lower level by KKT (KKT-1)–(KKT-3); keep each firm's price choice (or its log-price) as an upper variable and enforce the first-order condition (55)/(56) or let the solver find prices by maximizing the firm's profit with KKT embedded. This yields an MPCC.
- **COOP:** Replace the common lower level by KKT and maximize the coalition's objective; prices can be obtained either from (68)–(69) or as upper variables in the same MPCC. This is a single-leader MPEC.
- **HYB:** Embed each firm's lower-level KKT and both firms' price vectors; the Stage-1 game becomes an EPEC. Under Rosen's condition (diagonal dominance) we have uniqueness, and you may also solve by best responses using the closed-form Lerner (73) with the KKT providing μ^{HYB}.

Complementarity handling. Keep orthogonality $a \perp b$directly and use an MCP solver (PATH), or linearize with Fortuny-Amat–McCarl binaries $\zeta$and tight big-M bounds:

$$0 \leq a \leq M^a \zeta, 0 \leq b \leq M^b(1-\zeta). \tag{KKT.14}$$

Choose Ms from physical maxima (capacities, costs) and scale rows so $\| g \|_\infty \leq 1$.

Appendix C. Detailed Case Study Data and Parameter Calibration

The appendix reports detailed firm-specific information on production facilities, distribution centers, supplier pools, transportation costs and emissions, carbon policy parameters, and investment options for the two competing firms operating in Türkiye. Tables C.1–C.3 summarize, respectively, the firm-specific parameters for Firm 1, Firm 2, and the system-wide market, transportation, and policy parameters shared by both firms.

Table C1. Firm 1 (Türkiye) – Model parameters and calibrated data

Category	Parameter	Symbol	Firm 1 value	Unit	Notes
Production	Unit production cost (A/B)	$\left(c_{1jp}^{prod}\right)$	P1 Istanbul: 42 / 48; P2 Ankara: 44 / 50; P3 Gaziantep: 43 / 49	TRY/unit	Case data
	Nominal production capacity (A/B)	$\left(\overline{Q_{1jp}^{prod}}\right)$	P1: 28k / 22k; P2: 24k / 20k; P3: 20k / 18k	units/period	Case data
	Expandable capacity (A/B)	$\left(\Delta\overline{Q_{1jp}^{prod}}\right)$	P1: +8k / +6k; P2: +7k / +5k; P3: +6k / +5k	units/period	Discrete option
	Capacity expansion CAPEX	$\left(K_{1jp}^{cap}\right)$	P1: 3.2; P2: 2.8; P3: 2.2	M TRY	Case data
	Green retrofit emission reduction	$\left(\rho_{1jp}^{prod}\right)$	0.28; 0.25; 0.27	–	Case data
	Green retrofit investment	$\left(K_{1jp}^{green}\right)$	4.1; 3.6; 3.1	M TRY	Case data
	Production emission factor (pre-retrofit)	$\left(e_{1jp}^{prod}\right)$	Calibrated: A = 1.00, B = 1.08	kg CO_2/unit	Typical light manufacturing in TR; reduced by $\left((1-\rho)\right)$
Private DCs	DC handling capacity (A/B)	$\left(\overline{Q_{1kp}^{DC}}\right)$	D1 Istanbul: 30k / 24k; D2 Adana: 24k / 20k; D3 Diyarbakır: 18k / 15k	units/period	Case data
	Inventory holding cost	$\left(c_{1kp}^{Inv}\right)$	1.25; 1.10; 1.05	TRY/unit·period	Case data

Shared green DC	Pooled handling capacity	$\left(\overline{Q_{k'}^{DC,g}}\right)$	G1 Konya: 40,000	units/period	Case data
	Holding/handling cost	$\left(c_{1k'p}^{Inv}\right)$	1.40	TRY/unit·period	Case data
	Transport emission reduction	$\left(\rho_{k'}^{Trans}\right)$	0.35	–	Adjacent legs × 0.65
	Fixed activation cost	$\left(F_{k'}^{g}\right)$	6.0	M TRY	Shared via $\left(\lambda_{fk'}\right)$
Suppliers					
	Domestic purchase price (A/B)	$\left(c_{iat}^{wh}\right)$	17–19 / 20–23	TRY/unit	Ex-works
	Import purchase price (A/B)	$\left(c_{iat}^{wh}\right)$	15–16 / 18–19	EUR/unit	FX & tariffs applied
	Import tariff	(τ_i)	0.03–0.07	fraction	Case data
	Supplier capacity	(S_{iat}^{max})	16k–26k	units/period	Case data
	Lead time	(L_{ait})	2–3 (dom.), 5–7 (int.)	days	Case data
	Quality score	(q_{ai})	0.78–0.86	–	Case data
	Supplier emission factor	$\left(e_{ia}^{sup}\right)$	0.20–0.45	kg CO_2/unit	Case data
	Green supplier reduction	$\left(\rho_{ia}^{sup}\right)$	0.15	–	Case data
Policy & finance	Domestic sourcing floor (primary/secondary)	$\left(\rho_{1a}^{dom}\right)$	0.55 / 0.40	–	Case data
	Plant×product activation cost	$\left(F_{1jp}^{PL}\right)$	0.9–1.4	M TRY/period	Case data
	DC activation cost	$\left(F_{1k}^{DC}\right)$	0.8–1.2	M TRY/period	Case data
	Investment budget	(B_{1t})	18	M TRY/period	Case data
	Minimum price (margin floor)	(M_{1mct})	Cost + 8–10 %	TRY/unit	Policy constraint

Table C2. Firm 2 (Türkiye) – Model parameters and calibrated data

Category	Parameter	Symbol	Firm 2 value	Unit	Notes
Production	Unit production cost (A/B)	$\left(c_{1jp}^{prod}\right)$	P4 Izmir: 43 / 47; P5 Bursa: 45 / 51	TRY/unit	Case data
	Nominal production capacity (A/B)	$\left(\overline{Q_{1jp}^{prod}}\right)$	P4: 25k / 21k; P5: 22k / 19k	units/period	Case data
	Expandable capacity (A/B)	$\left(\Delta\overline{Q_{1jp}^{prod}}\right)$	P4: +7k / +5k; P5: +6k / +5k	units/period	Discrete option
	Capacity expansion CAPEX	$\left(K_{1jp}^{cap}\right)$	3.0; 2.5	M TRY	Case data

	Green retrofit emission reduction	$\left(\rho_{1jp}^{prod}\right)$	0.26; 0.24	–	Case data
	Green retrofit investment	$\left(K_{1jp}^{green}\right)$	3.9; 3.3	M TRY	Case data
	Production emission factor (pre-retrofit)	$\left(e_{1jp}^{prod}\right)$	Calibrated: A = 0.98, B = 1.06	kg CO_2/unit	Slightly lower due to coastal logistics
Private DCs	DC handling capacity (A/B)	$\left(\overline{Q_{1kp}^{DC}}\right)$	D4 Izmir: 26k / 22k; D5 Ankara: 20k / 18k	units/period	Case data
	Inventory holding cost	$\left(c_{1kp}^{Inv}\right)$	1.20; 1.15	TRY/unit·period	Case data
Shared green DC	Pooled capacity	$\left(\overline{Q_{k'}^{DC,g}}\right)$	G1 Konya: 40,000	units/period	Shared
	Holding/handling cost	$\left(c_{1k'p}^{Inv}\right)$	1.40	TRY/unit·period	Case data
	Transport emission reduction	$\left(\rho_{k'}^{Trans}\right)$	0.35	–	Adjacent legs × 0.65
	Cost-sharing weight	$\left(F_{k'}^{g}\right)$	Baseline: 0.50	–	Equal share benchmark
Suppliers					
	Domestic purchase price (A/B)	$\left(c_{iat}^{wh}\right)$	18–19 / 21–23	TRY/unit	Case data
	Import purchase price (A/B)	$\left(c_{iat}^{wh}\right)$	15–17 / 18–20	EUR/unit	Case data
	Import tariff	(τ_i)	0.03–0.07	fraction	Case data
	Supplier capacity	(S_{iat}^{max})	18k–22k	units/period	Case data
	Lead time	(L_{ait})	2–3 (dom.), 6–7 (int.)	days	Case data
	Quality score	(q_{ai})	0.79–0.85	–	Case data
Policy & finance	Domestic sourcing floor (primary/secondary)	$\left(e_{ia}^{sup}\right)$	0.50 / 0.40	–	Case data
	Plant×product activation cost	$\left(\rho_{ia}^{sup}\right)$	0.9–1.4	M TRY/period	Case data
	DC activation cost	$\left(\rho_{1a}^{dom}\right)$	0.8–1.2	M TRY/period	Case data
	Investment budget	$\left(F_{1jp}^{PL}\right)$	16	M TRY/period	Case data
	Minimum price (margin floor)	$\left(F_{1k}^{DC}\right)$	Cost + 8–10 %	TRY/unit	Policy constraint

Table C3. System-wide and market parameters (common to both firms)

Category	**Parameter**	**Symbol**	**Value**	**Unit**	**Notes (Türkiye-realistic)**

Demand	Base demand scale	(Θ_{mct})	TR-UR: 14k/9k; TR-UE: 12.5k/8.5k; TR-RR: 8k/5k; DE-EC: 9.5k/6k; ME-R: 6.5k/4.5k	units/period	Case data
	Own-price elasticity	(β_{mct})	1.35–1.85 (by market/channel)	–	Diagonal dominance preserved
	Cross-price elasticity	$\left(\eta_{mct}^{ff'}\right)$	0.20–0.30	–	Symmetric
	Brand/service intercept	(α_0)	0.82	–	Case data
	Service sensitivity	(α_1, α_2)	0.30; 0.08	–	Case data
	Mode penalty	(θ_v)	Truck 0.08; Rail 0.04; Ship 0.06	–	Reliability effect
	Service penalty weight	(O_{mct})	TR 4.0; DE 5.5; ME 4.5	TRY	Case data
Transport costs	Unit transport cost	(c_{vuht}^{Tran})	Truck 2–7; Rail 2–5; Ship 1.8–3.8	TRY/unit	Türkiye/EU corridors
Transport emissions	Mode emission factor	(e_{vuh}^{Tran})	Truck 0.26; Rail 0.08–0.10; Ship 0.05–0.07	kg CO_2/unit·leg	IPCC-consistent
Transport capacity	Mode capacity	(Cap_{vt}^{tran})	Truck 60k; Rail 45k; Ship 35k	units/period	Case data
Inventory	Spoilage rate	(ϕ)	0.1 %	%/period	Negligible
Carbon policy	Carbon tax (baseline)	(τ_t)	1,200	TRY/tCO_2	Case value
	Carbon permit price (baseline)	(π_t)	1,300	TRY/tCO_2	Case value
	Emission quota	$(\widehat{E_{ft}})$	Firm 1: 120; Firm 2: 110	tCO_2/period	Case data
	Trading bounds	$\left(v_{ft}^{min}, v_{ft}^{max}\right)$	−40, +60	tCO_2/period	Case data
Pricing rules	Price bounds	$\left(\underline{p}_{fmct}, \overline{p}_{fmct}\right)$	As specified per market/channel	TRY/unit	Case data
	Max price change	(ΔP_{mc}^{max})	7–10	TRY/period	Case data
	Channel price gap	(Δf_{mtch})	6 (dom.), 8 (export)	TRY	Case data

5.2. Computational Certification, Robustness, and ε-Equilibrium Verification

We now summarize the computational certification properties of the proposed solution approach and assess its robustness with respect to initialization, trust-region design, and numerical equilibrium accuracy. This subsection complements the case study results by formally documenting optimality guarantees, computational stability, and equilibrium verification checks.

Optimality gaps and computational performance

Across all scenarios NC, HYB, and COOP the proposed framework delivers solutions with certified near-optimality. As reported in Tables D1–D3, the strategic master problem terminates with optimality gaps consistently below 0.25%, while the baseline monolithic formulation exhibits substantially larger gaps (≈1%). Wall-clock times are decomposed into upper-level (strategic master) and lower-level (operational oracle) components, reflecting the bilevel structure of the problem. Median wall-times across multiple seeds remain within practical limits for all regimes, demonstrating the scalability of the proposed decomposition with equilibrium oracles. Importantly, the tighter gaps achieved by the proposed approach coincide with *lower* computational times relative to the baseline, confirming that improved solution quality is not obtained at the expense of computational burden.

Robustness to initialization

Robustness with respect to initialization is assessed by solving each scenario under multiple random seeds and reporting median outcomes and dispersion measures (stability columns in Tables D1–D3). Across all regimes, dispersion in profit, residuals, and solve times remains below 2%, and the economic ranking of governance regimes (HYB ≥ COOP ≥ NC in profits, COOP ≤ HYB ≤ NC in emissions) is invariant. This confirms that results are not driven by favorable starting points or path-dependent artifacts of the solution process.

Trust-region design and algorithmic sensitivity

The strategic master problem employs an adaptive binary trust-region, with a dynamically adjusted Hamming-ball radius bounded within $[2, 12]$. This mechanism stabilizes exploration of discrete design decisions without restricting convergence to high-quality solutions. Algorithmic sensitivity is examined through an ablation study (Table D15), comparing performance with and without DCEC. While the inclusion of DCEC substantially reduces the number of master iterations and wall-time and tightens final optimality gaps, solution values and regime rankings remain unchanged. This demonstrates that the reported results are robust to trust-region settings and cut management strategies and do not rely on fine-tuned parameter choices.

ε-equilibrium definition and verification

For the NC and HYB regimes, strategic pricing is determined through mixed complementarity formulations. Solutions are therefore delivered as ε-equilibria, where ε is defined as the maximum unilateral profit deviation across all firms and market–channel tuples, computed via the MCP natural residual. $\varepsilon = \max\{\| F(z) \|_\infty\}$, with $F(z)$denoting the MCP residual mapping. Across all reported instances, ε remains below 10^{-6}, as documented in Tables D1–D3. For the COOP regime, equilibrium conditions are verified through KKT residuals of the centralized pricing problem, which are likewise below 10^{-6} at termination.

Additional verification checks include: (i) primal and dual feasibility of all lower-level LPs within 10^{-8} tolerances, (ii) consistency between realized flows and market-clearing conditions, and (iii) stability of objective values and multipliers upon re-solving with alternative solvers (IPOPT vs. PATH for pricing oracles, CPLEX vs. Gurobi for LPs; see Table D15). Together, these checks ensure that no firm can unilaterally deviate to improve its payoff beyond numerical tolerance, and that reported outcomes represent economically meaningful equilibria rather than numerical artifacts.

Table D15. Algorithmic ablation and Benchmark Comparison				
Configuration	Master iters	Time (min)	Final gap (%)	UB difference
Standard LBB (single optimality cuts)	28	9.6	0.92	−1.14 M TRY
LBB without equilibrium oracles	24	8.1	0.71	−0.78 M TRY
LBB-EO (no DCEC)	19	6.8	0.48	−0.23 M TRY
LBB-EO + DCEC (proposed)	10	4.1	0.21	baseline
COOP price oracle: IPOPT	—	2.2	KKT 7×10^{-7}	same UB
COOP price oracle: PATH-NLP	—	2.9	KKT 8×10^{-7}	same UB
Follower LP: CPLEX vs Gurobi	—	1.0 vs 1.05	—	same UB

5.4. Sensitivity Analysis

Table D16 in Appendix D extends robustness to the calibration levers explicitly service proxy parameters (α_2, θ_v) and procurement scoring weights $(\lambda_\ell, \lambda_e)$, and confirms that HYB/COOP remain more responsive than NC because they can exploit pooling, mode shifting, and the shared green corridor, yielding larger profit gains and larger emission reductions for the same parameter shift. Finally, Table D17 in Appendix D provides ablation checks that rule out a double-counting interpretation λ_e acts as a procurement screening preference, whereas τcap-and-trade price realized emissions; switching each channel on/off produces distinct and interpretable changes in profit and emissions, while preserving qualitative scenario insights.

The results reported in Table D16 confirm that the main economic–environmental trade-offs of the model are robust to simultaneous perturbations of the parameters $(\alpha, \theta, \lambda)$. Increasing the service–price coupling parameters $\alpha_2$and $\theta_v$leads to systematic profit improvements under HYB and COOP, while NC exhibits only marginal responses. Quantitatively, HYB shows the largest profit sensitivity, reflecting its ability to monetize service improvements through independent pricing, whereas COOP translates the same changes primarily into additional abatement rather than profit. Variations in the carbon- and lead-time–related weights $(\lambda_e, \lambda_\ell)$generate limited profit volatility (sub-1% across all regimes) but induce material emission reductions, especially under COOP, where centralized routing and pricing compress the shadow cost of the emission constraint. HYB remains close to COOP in emission response but preserves a small profit advantage, consistent with the baseline findings. Importantly, no parameter realization reverses the regime ranking HYB ≥ COOP ≥ NCin profit, nor does any perturbation induce unstable oscillations or binding conflicts between carbon tax and quota mechanisms. These results demonstrate that $(\alpha, \theta, \lambda)$primarily scale the strength of observed mechanisms service monetization, pooled routing, and permit banking without altering their direction, confirming the structural robustness of the proposed formulation to calibration uncertainty in critical parameters.

Table D17 shows when $\lambda_e$is switched off, profits rise marginally but emissions deteriorate, showing that supplier screening contributes incremental abatement beyond the priced-carbon mechanism. Conversely, removing priced carbon (τ and cap/trading) increases profits but increases emissions substantially, indicating that compliance instruments are the primary driver of monetized abatement. Across both ablations, the qualitative scenario ranking and the operational mechanisms (mode shifting and hub consolidation) remain stable, supporting robustness of the main conclusions.

Table D16. Parametric Sensitivity of Profit and Emissions to Service, Demand, and Carbon Cost Couplings

Targeted check	Meaning	NC: ΔProfit / ΔEmis	HYB: ΔProfit / ΔEmis	COOP: ΔProfit / ΔEmis
$(\boldsymbol{\alpha_2 + 15\%})$	stronger mode-quality impact on service	+0.10 / −0.20	+0.35 / −0.60	+0.30 / −0.70
$(\boldsymbol{\theta_{rail} + 20\%})$	rail seen as more reliable → service improves when shifting to rail	+0.05 / −0.10	+0.25 / −0.45	+0.22 / −0.55
$(\boldsymbol{\theta_{ship} + 20\%})$	ship reliability weight up on export lanes	+0.04 / −0.08	+0.18 / −0.35	+0.16 / −0.40
Re-weight score toward lead time:$(\boldsymbol{\lambda_l} \uparrow)(\boldsymbol{renorm})$	awards favor faster suppliers	+0.03 / −0.10	+0.10 / −0.25	+0.08 / −0.22
Re-weight score toward supplier carbon: $(\boldsymbol{\lambda_e} \uparrow)(\boldsymbol{renorm})$	awards favor lower $\left(e_{ia}^{sup}\right)$ suppliers	−0.02 / −0.18	−0.05 / −0.45	−0.06 / −0.55

Table D17. System Performance Under Combined Market, Capacity, and Carbon Policy Shocks

Ablation	What it isolates	NC: ΔProfit / ΔEmis	HYB: ΔProfit / ΔEmis	COOP: ΔProfit / ΔEmis
$\boldsymbol{Set(\lambda_e = 0)\left(keep\left(\tau, \hat{E}, \pi\right)active\right)}$	removes *screening* channel, keeps priced carbon	+0.03 / +0.10	+0.06 / +0.25	+0.05 / +0.30
$\boldsymbol{Set(\tau = 0)andremovecap/trading\left(keep(\lambda_e > 0)\right)}$	removes *priced carbon*, keeps screening	+0.40 / +1.10	+0.55 / +1.40	+0.50 / +1.60

References

Al-Dujaili, A., Srikant, S., Hemberg, E., & O'Reilly, U. M. (2019, February). On the application of Danskin's theorem to derivative-free minimax problems. In AIP Conference Proceedings (Vol. 2070, No. 1, p. 020026). AIP Publishing LLC.

Allende, G. B., & Still, G. (2013). Solving bilevel programs with the KKT-approach. Mathematical Programming, 138(1), 309-332.

Bai, Y., Li, H., & Gu, M. (2024). Advance selling policies with stochastic consumer valuations and advertising effects: dynamic pricing and inventory decisions. International Journal of Production Economics, 277, 109401.

Berge, C. (1984). Hypergraphs: combinatorics of finite sets (Vol. 45). Elsevier.

European Commission. (2023). *EU Emissions Trading System (EU ETS)*. Brussels: European Commission. https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets_en

Gholami, R. A., Sandal, L. K., & Ubøe, J. (2021). A solution algorithm for multi-period bi-level channel optimization with dynamic price-dependent stochastic demand. Omega, 102, 102297.

Glicksberg, I. L. (1952). A further generalization of the Kakutani fixed point theorem, with application to Nash equilibrium points. Proceedings of the American Mathematical Society, 3(1), 170-174.

Guajardo, M., & Rönnqvist, M. (2015). Operations research models for coalition structure in collaborative logistics. European Journal of Operational Research, 240(1), 147-159.

Han, B., Chi, T., Hu, F., & Wang, M. (2025). Pricing decisions and fairness concerns: supply chain power structures. Management Decision, 63(7), 2105-2124.

Israel, B., & Siwandeti, M. (2024). Agri-food supply chain capabilities and smallholders' on-farm economic viability: a moderated mediation analysis of subsidy schemes and dynamic pricing. Cogent Food & Agriculture, 10(1), 2391565.

International Energy Agency (IEA). (2024). *CO_2 emissions in 2023*. Paris: International Energy Agency. https://www.iea.org/reports/co2-emissions-in-2023

International Energy Agency (IEA). (2023). *Road transport*. Paris: International Energy Agency. https://www.iea.org/reports/road-transport

Kakutani, S. (1941). A generalization of Brouwer's fixed point theorem.

Karush, W. (1939). Minima of functions of several variables with inequalities as side constraints. M. Sc. Dissertation. Dept. of Mathematics, Univ. of Chicago.

Kjeldsen, T. H. (2000). A contextualized historical analysis of the Kuhn–Tucker theorem in nonlinear programming: the impact of World War II. Historia Mathematica, 27(4), 331-361.

Kuhn, H. W., & Tucker, A. W. (2013). Nonlinear programming. In Traces and emergence of nonlinear programming (pp. 247-258). Basel: Springer Basel.

Liu, X., Li, J., Wu, J., & Zhang, G. (2017). Coordination of supply chain with a dominant retailer under government price regulation by revenue sharing contracts. Annals of Operations Research, 257(1), 587-612

Li, Y., Li, X., & Zhang, S. (2021). Optimal pricing of customized bus services and ride-sharing based on a competitive game model. Omega, 103, 102413.

Li, Y., Xiong, Y., & Zhou, Y. (2023). Dynamic pricing strategies: Uniform or discrimination for multi-channel retailing under platform-based supply chain competition. Journal of Business Research, 166, 114087.

Lin, X., & Wang, H. (2022). Dynamic pricing for online information services considering service duration and quality level. Omega, 109, 102593.

Lerner, A. P. (1934). The concept of monopoly and the measurement of monopoly power. The Review of Economic Studies, 1(3), 157-174.

Magro-Montero, M., Santarsiero, F., & Morales-Alonso, G. (2025). Value creation and capture in innovation ecosystems: strategic cooperation between e-commerce platforms and third-party logistics (3PLs) providers. European Journal of Innovation Management, 1-24.

McKinsey & Company. (2023). *Decarbonizing logistics: Charting the path ahead*. McKinsey & Company. https://www.mckinsey.com/capabilities/operations/our-insights/decarbonizing-logistics-charting-the-path-ahead

Rahmani, D., & Pashapour, A. (2024). Dynamic pricing decision for new and returned products in a dual-channel supply chain based on customer segmentation. Soft Computing, 28(23), 13205-13224.

Samson, O. J., Awotundun, T. A., Faniyi, T. O., & Olabiwoninu, A. A. (2026). Multinational Enterprises and Sustainability of Agri-Food Supply Chains. In Global Food Value Chains: An In-depth Exploration of International Trade Dynamics (pp. 129-146). Cham: Springer Nature Switzerland.

Stone, M. H. (1948). The generalized Weierstrass approximation theorem. Mathematics Magazine, 21(5), 237-254.

Varma, S., Patra, P., & Shankar, R. (2025). The food industry supply chain: Synergy of environmental orientation and collaboration in implementing circularity. Business Strategy and the Environment, 34(1), 218-241.

Wang, X., & Du, S. (2025). Strategic pricing and quality discrimination in presence of different customer incentive programs. Omega, 103336.

World Economic Forum. (2023). *Net-Zero Industry Tracker 2023 Edition*. Geneva: World Economic Forum. https://www.weforum.org/publications/net-zero-industry-tracker-2023/

Xin, C., Zhou, Y., Sun, M., & Chen, X. (2022). Strategic inventory and dynamic pricing for a two-echelon green product supply chain. Journal of Cleaner Production, 363, 132422.

Yan, L., Hong, P., & Wu, Z. (2024). Dynamic pricing and emission reduction efforts in a dual-channel green supply chain under bidirectional free riding. Journal of Cleaner Production, 438, 140713.

Zhou, Y., Mandania, R., & Liu, J. (2022). Green vehicle routing and dynamic pricing for scheduling on-site services. International Journal of Production Economics, 254, 108602.